

UNIT -IV

CAPITAL BUDGETING

Capital is defined as wealth, which is created over a period of time through abstinence to spend. There are different forms of capital property, cash or titles to wealth. It is the aggregate of funds used in the short run and long run. An economist views capital as the value total assets available with the business. An accountant sees the capital as the difference between the assets and liabilities.

Significance of capital

1. **To promote a business:** Capital is required at the promotion stage. A large variety of expenses have to be incurred on project reports, feasibility studies and reports, preparation and filing of various documents, and for meeting various other expenses in connection with the raising of capital from the public.
2. **To conduct business operations smoothly:** Business firms also need capital for the purpose of conducting their business operations such as research and development, advertising, sales promotion, distribution and operation expenses.
3. **To expand and diversify:** The firm requires a lot of capital for expansion and diversification purposes. This includes development expense such as purchase of sophisticated machinery and equipment and also payment towards sophisticated technology.
4. **To meet contingencies:** A firm needs funds to meet contingencies such as sudden fall in sales, major litigation, nature calamities like fire, and so on.
5. **To pay taxes:** The firm has to meet its statutory commitments such as income tax and sales tax, excise duty and so on.
6. **To pay dividends and interests:** The business has to make payment towards dividends and its interest to shareholders and financial institutions respectively.
7. **To replace the assets:** The business needs to replace its assets like plant and machinery after a certain period of use. For this purpose the firm needs funds to make suitable replacement of assets in place of old and worn out assets.
8. **To support welfare programmes:** The company may also have to take up social welfare programmes such as literacy drive, and health camps, It may have to donate to charitable trusts, educational institutions or public services organizations.
9. **To wind up:** At the time of winding up, the company may need funds to meet liquidation expenses

Types of capital

- A) Fixed capital
- B) Working capital

FIXED CAPITAL

Fixed capital is that portion of capital which invested in acquiring long term assets such as land and buildings, plant and machinery, furniture and fixtures, and so on, fixed capital forms the skeleton of the business. It provides the basic assets as per the business needs.

Features of fixed assets:

1. **Permanent in nature:** fixed capital is more or less permanent in nature, it is generally not withdrawn as long as the business carries on its business.
2. **Profit generation:** fixed asset are the sources of profits but they can never generate profits by themselves. They use stocks, cash and debtors to generate profits.
3. **Low liquidity:** the fixed assets cannot be converted into cash quickly. Liquidity refers to conversion of assets into cash.
4. **Amount of fixed capital :** the amount of fixed capital of a company depends on a number of factors such as size of the company, nature of business, method of production and so on. A manufacturing company such as steel factory may require relatively large finance when compared to a service organization such as a software company.
5. **Utilized for promotional and expansion:** the fixed capital is mostly needed at the time of promoting the company to purchase the fixed assets or at the time of expansion. In other words, the need for fixed capital arises less frequently.

Types of fixed assets

1. **Tangible fixed assets :** these are physical items which can be seen and touched. Most of the common fixed assets are land, buildings, machinery, motor vehicles, furniture and so on.
2. **Intangible fixed assets :** these do not have physical form. They cannot be seen or touched. But these are very valuable to business. Examples are goodwill, brand names, trademarks, patents, copy rights and so on.
3. **Financial fixed assets :** these are investments in shares, foreign currency deposits, government bonds , shares held by the business in other companies and so on.

WORKING CAPITAL

Working capital is the flesh and blood of the business. It is that portion of capital that makes a company work. It is not just possible to carry on the business with only fixed assets. Working capital is a must, working capital is also called circulating capital. It is used to meet regular or recurring needs of the business. The regular needs refer to the purchase of materials, payment of wages and salaries, expenses like rent, advertising, power and so on. In short , working capital is the amounts needed to cover the cost of operating the business.

Definition of working capital

Working capital define as a current assets excess of current liabilities

Its also define in mathematically formula as

Working capital = current assets – current liabilities features of working capital

1. Short life span: working capital changes in its form cash to stock, stock to debtors, debtors to cash, the cash balances may be kept idle for a week or so, debtors have a life span of a few months , raw materials are held for a short – time until they go into production, finished goods asheld for a short – time until they are sold.
2. Smoothly flow of operations: adequate amount of working capital enables the business to conduct its operations smoothly. It is there fore, called the flesh and blood of the business.
3. Liquidity: the assets represented by the working capital can be converted into cash quicklywithin a short period of time unlike fixed assets.
4. Amount of working capital: the amount of working capital of a business depends on many factors such as size and nature of the business, production and marketing policies, business cycles and so on.
5. Utilized for payment of current expenses: the working capital is used to pay for current expenses such as suppliers of raw materials, payment of wages and salaries, rent and other expenses andso on.

Components of working capital:

Current assets: current assets are those assets which are converted into cash with in accounting period or within the year. For example, cash in hand, cash at bank, sundry debtor, bill receivable, prepaid expenses etc.

Current liabilities: current liabilities are those liabilities to pay outside with in the year. For example sundry creditor, bill payable, bank overdraft, outstanding expenses.

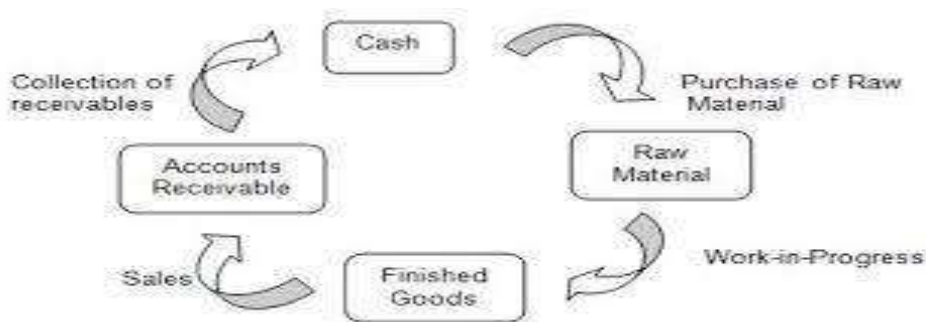
Gross working capital:

In the broader sense, the term working capital refers to the gross working capital. The notion of the gross working capital refers to the capital invested in total current assets of the enterprise. Current assets are those assets, which in the ordinary course of business, can be converted into cash within a short period, normally one accounting year.

Net working capital:

In a narrow sense, the term working capital refers to the net working capital. Networking capital represents the excess of current assets over current liabilities.

WORKING CAPITAL CYCLE



Factors determining the working capital requirements

1. **Nature or character of business:** The working capital requirements of a firm basically depend upon the nature of its business. Public utility undertakings like electricity, water supply and railways need very limited working capital as their sales are on cash and are engaged in provision of services only. On the other hand, trading firms require more investment in inventories, receivables and cash and such they need large amount of working capital. The manufacturing undertakings also require sizable working capital.
2. **Size of business or scale of operations:** The working capital requirements of a concern are directly influenced by the size of its business, which may be measured in terms of scale of operations. Greater the size of a business unit, generally, larger will be the requirements of working capital. However, in some cases, even a smaller concern may need more working capital due to high overhead charges, inefficient use of available resources and other economic disadvantages of small size.
3. **Production policy:** If the demand for a given product is subject to wide fluctuations due to seasonal variations, the requirements of working capital, in such cases, depend upon the production policy. The production could be kept either steady by accumulating inventories during slack periods with a view to meet high demand during the peak season or the production could be curtailed during the slack season and increased during the peak season. If the policy is to keep the production steady by accumulating inventories it will require higher working capital.
4. **Manufacturing process/Length of production cycle:** In manufacturing business, the requirements of working capital will be in direct proportion to the length of manufacturing process. Longer the process period of manufacture, larger is the amount of working capital required, as the raw materials and other supplies have to be carried for a longer period.
5. **Seasonal variations:** If the raw material availability is seasonal, they have to be bought in bulk during the season to ensure an uninterrupted material for the production. A huge amount is, thus, blocked in the form of material, inventories during such season, which give rise to more working capital requirements. Generally, during the busy season, a firm

requires larger working capital than in the slack season.

6. **Working capital cycle:** In a manufacturing concern, the working capital cycle starts with the purchase of raw material and ends with the realization of cash from the sale of finished products. This cycle involves purchase of raw materials and stores, its conversion into stocks of finished goods through work-in progress with progressive increment of labour and service costs, conversion of finished stock into sales, debtors and receivables and ultimately realization of cash. This cycle continues again from cash to purchase of raw materials and so on. In general the longer the operating cycle, the larger the requirement of working capital.
7. **Credit policy:** The credit policy of a concern in its dealings with debtors and creditors influences considerably the requirements of working capital. A concern that purchases its requirements on credit requires lesser amount of working capital compared to the firm, which buys on cash. On the other hand, a concern allowing credit to its customers shall need larger amount of working capital compared to a firm selling only on cash.
8. **Business cycles:** Business cycle refers to alternate expansion and contraction in general business activity. In a period of boom, i.e., when the business is prosperous, there is a need for larger amount of working capital due to increase in sales. On the contrary, in the times of depression, i.e., when there is a down swing of the cycle, the business contracts, sales decline, difficulties are faced in collection from debtors and firms may have to hold large amount of working capital.
9. **Rate of growth of business:** The working capital requirements of a concern increase with the growth and expansion of its business activities. The retained profits may provide for a part of working capital but the fast growing concerns need larger amount of working capital than the amount of undistributed profits.

METHODS AND SOURCES OF FINANCE/ CAPITAL

Methods of finance

1. Long term finance
2. Medium term finance
3. Short term finance

SOURCES OF FINANCE

1. **Long term finance:** long term finance available for a long period say five years and above. The long term methods outlined below are used to purchase fixed assets such as land and buildings, plant and so on.
 - a) **Own capital :** irrespective of the form of organization such as soletrader, partnership or a company, the owners of the business have to invest their own finances to start with. Money invested by the owners, partners or promoters is permanent and will stay with the business throughout the life of business.

- b) **Share capital** : normally in the case of a company, the capital is raised by issue of shares. The capital so raised is called share capital. The share capital can be of two types, preference share capital and equity share capital.
- c) **Debentures**: debentures are the loans taken by the company. It is a certificate or letter by the company under its common seal acknowledging the receipt of loan. A debenture holder is the creditor of the company. A debenture holder is entitled to a fixed rate of interest on the debenture amount.
- d) **Government grants and loans**: government may provide long term finance directly to the business houses or by indirectly subscribing to the shares of the companies. The government gives loans only if the project satisfies certain conditions, such as setting up a project in a notified area, or ventures into projects which are beneficial for the society as a whole.

2. Medium term finance

- a. Bank loans ; bank loans are extended at a fixed rate of interest. Repayment of the loan and interest are scheduled at the beginning and are usually directly debited to the current account of the borrower. These are secured loans.
- b. Hire purchase: it is a facility to buy a fixed asset while paying the price over a long period of time. In other words , the possession of the asset can be taken by making a down payment of a part of the price and the balance will be repaid with a fixed rate of interest in agreed number of installments.
- c. Leasing or renting: where there is a need for fixed assets, the asset need not be purchased. It can be taken on lease or rent for specified number of years. The company who owns the assets is called lessor and the company which takes the asset on lease is called lessee. The agreement between the lessor and lessee is called a lease agreement.
- d. Venture capital: this form of finance is available only for limited companies. Venture capital is normally provided in such projects where there is relatively a higher degree of risk. For such projects, finance through the conventional sources may not be available. Many banks offer such finance through their merchant banking divisions, or specialist banks which offer advice and financial assistance. The financial assistance may take form of loans and venture capital.

3. SHORT TERM FINANCE

- a. Commercial paper: it is new money market instrument introduced in india in recent times. Cps are issued in large denominations by the leading, nationally reputed, highly rated and credit worthy, large manufacturing and finance companies in the public and private sector. The proceeds of the issue of commercial paper are used to finance current transactions and seasonal and interim needs for funds.
- b. Bank overdraft: this is special arrangement with the banker where the customer can draw more than what he has in his saving/ current account subject to a maximum limit. interest is charged on a day to day basis on the actual amount overdrawn .
- c. Trade credit: this is short term credit facility extended by the creditors to the debtors, normally, it is common for the traders to buy the materials and other supplies from the

suppliers on credit basis. After selling the stocks the traders pay the cash and buy fresh stocks again on credit. Sometimes , the suppliers may insist on the buyer to sign a bill.

CAPITAL BUDGETING

Capital budgeting is the process of making investment decision in long-term assets or courses of action. Capital expenditure incurred today is expected to bring its benefits over a period of time. These expenditures are related to the acquisition & improvement of fixes assets.

Capital budgeting is the planning of expenditure and the benefit, which spread over a number of years. It is the process of deciding whether or not to invest in a particular project, as the investment possibilities may not be rewarding. The manager has to choose a project, which gives a rate of return, which is more than the cost of financing the project. For this the manager has to evaluate the worth of the projects in-terms of cost and benefits. The benefits are the expected cash inflows from the project, which are discounted against a standard, generally the cost of capital.

Methods of capital budgeting

The capital budgeting appraisal methods are techniques of evaluation of investment proposal will help the company to decide upon the desirability of an investment proposal depending upon their; relative income generating capacity and rank them in order of their desirability. These methods provide the company a set of norms on the basis of which either it has to accept or reject the investment proposal. The most widely accepted techniques used in estimating the cost-returns of investment projects can be grouped under two categories.

1. Traditional methods
2. Discounted Cash flow methods

1. Traditional methods

These methods are based on the principles to determine the desirability of an investment project on the basis of its useful life and expected returns. These methods depend upon the accounting information available from the books of accounts of the company. These will not take into account the concept of „time value of money“, which is a significant factor to determine the desirability of a project in terms of present value.

A. Pay-back period method: It is the most popular and widely recognized traditional method of evaluating the investment proposals. It can be defined, as „the number of years required to recover the original cash out lay invested in a project“.

According to Weston & Brigham, “The pay back period is the number of years it takes the firm to recover its original investment by net returns before depreciation, but after taxes”.

According to James. C. Vanhorne, “The payback period is the number of years required to recover initialcash investment.

The pay back period is also called payout or payoff period. This period is calculated by dividing the cost of the project by the annual earnings after tax but before depreciation under this method the projects are ranked on the basis of the length of the payback period. A project with the shortest payback period will be given the highest rank and taken as the best investment.

Merits:

1. It is one of the earliest methods of evaluating the investment projects
2. It is simple to understand and to compute.
3. It does not involve any cost for computation of the payback period
4. It is one of the widely used methods in small scale industry sector
5. It can be computed on the basis of accounting information available from the books.

Demerits:

1. This method fails to take into account the cash flows received by the company after the payback period.
2. It doesn't take into account the interest factor involved in an investment outlay.
3. It doesn't take into account the interest factor involved in an investment outlay.
4. It is not consistent with the objective of maximizing the market value of the company's share.
5. It fails to consider the pattern of cash inflows i. e., the magnitude and timing of cash in flows.

**Cash outlay (or) original cost of
project**

Pay-back period

=

Annual cash inflow

B. Accounting (or) Average rate of return method (ARR):

It is an accounting method, which uses the accounting information repeated by the financial statements to measure the probability of an investment proposal. It can be determined by dividing the average income after taxes by the average investment i.e., the average book value after depreciation.

According to „Soloman“, accounting rate of return on an investment can be calculated as the ratio of accounting net income to the initial investment, i.e.,

$$\text{Average rate of return} = \frac{\text{average annual profit after tax}}{\text{Average investment}} \times 100$$

$$\text{Average annual profit after tax} = \frac{\text{sum of profit after tax}}{\text{No. of the years}}$$

$$\text{Average investment} = \frac{\text{cost} - \text{scrap value}}{\text{No. of the years}} + \text{additional working capital} + \text{scrap value}$$

Merits:

1. It is very simple to understand and calculate.
2. It can be readily computed with the help of the available accounting data.
3. It uses the entire stream of earning to calculate the ARR.

Demerits:

1. It is not based on cash flows generated by a project.
2. This method does not consider the objective of wealth maximization
3. IT ignores the length of the projects useful life.
4. It does not take into account the fact that the profits can be re-invested.

II: Discounted cash flow methods:

The traditional method does not take into consideration the time value of money. They give equal weight age to the present and future flow of incomes. The DCF methods are based on the concept that a rupee earned today is more worth than a rupee earned tomorrow. These methods take into consideration the profitability and also time value of money.

A. Net present value method (NPV)

The NPV takes into consideration the time value of money. The cash flows of different years are valued differently and made comparable in terms of present values for this the net cash inflows of various period are discounted using required rate of return which is predetermined.

According to Ezra Solomon, “It is a present value of future returns, discounted at the required rate of return minus the present value of the cost of the investment.”

NPV is the difference between the present value of cash inflows of a project and the initial cost of the project.

According to the NPV technique, only one project will be selected whose NPV is positive or above zero. If a project(s) NPV is less than „Zero“. It gives negative NPV hence. It must be rejected. If there are more than one project with positive NPV's the project is selected whose NPV is the highest.

NPV = PRESENT VALUE OF CASH INFLOW – PRESENT VALUE OF CASH OUTFLOW

Merits:

1. It recognizes the time value of money.
2. It is based on the entire cash flows generated during the useful life of the asset.
3. It is consistent with the objective of maximization of wealth of the owners. The ranking of projects is independent of the discount rate used for determining the present value.

Demerits:

1. It is difficult to understand and use.
2. The NPV is calculated by using the cost of capital as a discount rate. But the concept of cost of capital. If self is difficult to understand and determine.
3. It does not give solutions when the comparable projects are involved in different amounts of investment.

4. It does not give correct answer to a question whether alternative projects or limited funds are available with unequal lives.

Problem 1

The **cost** of a project is \$50,000 and it generates cash inflows of \$20,000, \$15,000, \$25,000, and \$10,000 over four years.

Required: Using the present value index method, appraise the profitability of the proposed investment, assuming a 10% rate of discount.

Solution

The first step is to calculate the present value and profitability index.

Year	Cash Inflows \$	Present Value Factor @10%	Present Value \$
1	20,000	0.909	18,180
2	15,000	0.826	12,390
3	25,000	0.751	18,775
4	10,000	0.683	6,830
			56,175

Total present value = \$56,175

Less: initial outlay = \$50,000

Net present value = \$6,175

Profitability Index (gross) = Present value of cash inflows / Initial cash outflow
= 56,175 / 50,000
= 1.1235

Given that the profitability index (PI) is greater than 1.0, we can accept the proposal.

Net Profitability = NPV / Initial cash outlay
= 6,175 / 50,000 = 0.1235
N.P.I. = 1.1235 – 1 = 0.1235

Given that the net profitability index (NPI) is positive, we can accept the proposal.

Problem 2

A company is considering whether to purchase a new machine. Machines A and B are available for \$80,000 each. Earnings after taxation are as follows:

Year	Machine A	Machine B
	\$	\$
1	24,000	8,000
2	32,000	24,000
3	40,000	32,000
4	24,000	48,000
5	16,000	32,000

Required: Evaluate the two alternatives using the following: (a) payback method, (b) rate of return on investment method, and (c) net present value method. You should use a discount rate of 10%.

Solution

(a) Payback method

24,000 of 40,000 = 2 years and 7.2 months

Payback period:

Machine A: $(24,000 + 32,000 + 1 \frac{3}{5} \text{ of } 40,000) = 2 \frac{3}{5}$ years.

Machine B: $(8,000 + 24,000 + 32,000 + 1/3 \text{ of } 48,000) = 3 \frac{1}{3}$ years.

According to the payback method, Machine A is preferred.

(b) Rate of return on investment method

Particular	Machine A	Machine B
Total Cash Flows	1,36,000	1,44,000
Average Annual Cash Flows	$1,36,000 / 5 = \$27,000$	$1,44,000 / 5 = \$28,800$
Annual Depreciation	$80,000 / 5 = \$16,000$	$80,000 / 5 = \$16,000$
Annual Net Savings	$27,200 - 16,000 = \$11,200$	$28,800 - 16,000 = \$12,800$

Average Investment	$80,000 / 2 = \$40,000$	$80,000 / 2 = \$40,000$
ROI = (Annual Net Savings / Average Investments) x 100	$(11,200 / 40,000) \times 100$	$(12,800 / 40,000) \times 100$
	= 28%	= 32%

According to the rate of return on investment (ROI) method, Machine B is preferred due to the higher ROI rate.

(c) Net present value method

The idea of this method is to calculate the present value of cash flows.

Year	Discount Factor	Machine A		Machine B	
	(at 10%)	Cash Flows (\$)	P.V (\$)	Cash Flows (\$)	P.V (\$)
1	.909	24,000	21,816	8,000	7,272
2	.826	32,000	26,432	24,000	19,824
3	.751	40,000	30,040	32,000	24,032
4	.683	24,000	16,392	48,000	32,784
5	.621	16,000	9,936	32,000	19,872
		1,36,000	1,04,616	1,44,000	1,03,784

Net Present Value = Present Value – Investment

Net Present Value of Machine A: $\$1,04,616 - \$80,000 = \$24,616$

Net Present Value of Machine B: $\$1,03,784 - 80,000 = \$23,784$

According to the net present value (NPV) method, Machine A is preferred because its NPV is greater than that of Machine B.

Problem 3

At the beginning of 2015, a business enterprise is trying to decide between two potential investments.

Required: Assuming a required rate of return of 10% p.a., evaluate the investment proposals under: (a) return on investment, (b) payback period, (c) discounted payback period, and (d) profitability index.

The forecast details are given below.

	Proposal A	Proposal B
Cost of Investment	\$20,000	28,000
Life	4 years	5 years
Scrap Value	Nil	Nil
Net Income (After depreciation and tax)		
End of 2015	\$500	Nil
End of 2016	\$2,000	\$3,400
End of 2017	\$3,500	\$3,400
End of 2018	\$2,500	\$3,400
End of 2019	Nil	\$3,400

It is estimated that each of the alternative projects will require an additional **working capital** of \$2,000, which will be received back in full after the end of each project.

Depreciation is provided using the straight line method. The **present value** of \$1.00 to be received at the end of each year (at 10% p.a.) is shown below:

Year	1	2	3	4	5
P.V.	0.91	0.83	0.75	0.68	0.62

Solution

Calculation of profit after tax

Year	Proposal A \$20,000			Proposal B \$28,000		
	Net Income	Dep.	Cash Inflow	Net Income	Dep.	Cash Inflow
	\$	\$	\$	\$	\$	\$
2015	500	5,000	5,500	—	5,600	5,600
2016	2,000	5,000	7,000	3,400	5,600	9,000
2017	3,500	5,000	8,500	3,400	5,600	9,000
2018	2,500	5,000	7,500	3,400	5,600	9,000
2019	—	—	—	3,400	5,600	9,000
Total	8,500	20,000	28,500	13,600	28,000	41,600

(a) Return on investment

	Proposal A	Proposal B
Investment	$20,000 + 2,000 = 22,000$	$28,000 + 2,000 = 30,000$
Life	4 years	5 years
Total Net Income	\$8,500	\$13,600
Average Return (\$)	$8,500 / 4 = 2,125$	$13,600 / 5 = 2,720$
Average Investment (\$)	$(22,000 + 2,000) / 2 = 12,000$	$(30,000 + 2,000) / 2 = 16,000$

Average Return on Average Investment (\$)	$(2,125 / 12,000) \times 100 = 17.7\%$	$(2,720 / 16,000) \times 100 = 17\%$
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(b) Payback period

Proposal A	Cash Inflow (\$)
2015	5,500
2016	7,000
2017	7,500 (7,500 / 8,500 = 0.9)
	20,000

Payback period = 2.9 years

Proposal B	Cash Inflow
	\$
2015	5,600
2016	9,000
2017	9,000
2018	4,400 (4,400 / 9,000 = 0.5)

Payback period = 3.5 years

(c) Discounted payback period

Proposal A		Proposal B	
P.V. of Cash Inflow		P.V. of Cash Inflow	
Year	\$	Year	\$
2015	5,005	2015	5,096
2016	5,810	2016	7,470
2017	6,375	2017	6,750
2018	2,810 (2,810 / 5,100 = 0.5)	2018	6,120
		2019	2,564 (2,564 / 5,580 = 0.4)
	20,000		28,000

Discounted Payback Period = 3.5 years	Discounted Payback Period = 4.4 years
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(d) Profitability index method

	Proposal A	Proposal B
Gross Profitability Index	$(22,290 / 20,000) \times 100$ = 111.45%	$(31,016 / 28,000) \times 100$ = 111.08%
Net Profitability Index	$(2,290 / 20,000) \times 100$ = 11.45%	$(3,016 / 28,000) \times 100$ = 10.8%